Module 8: Metabolism

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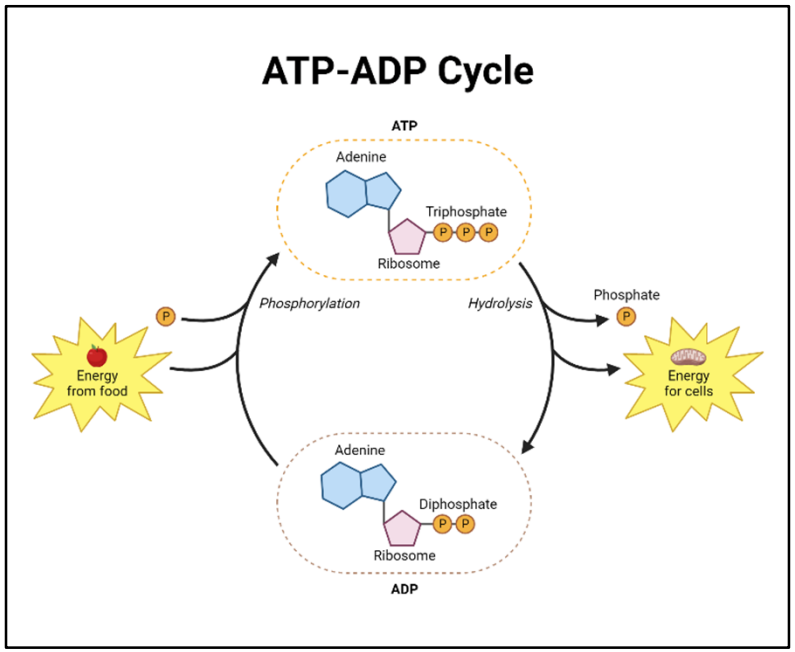
ATP Cycle

- **Adenosine 5'-diphosphate (ADP)** is converted to ATP by adding phosphate in an energetically unfavorable phosphorylation process.
- To make ADP and **inorganic phosphate (Pi),** ATP hydrolyzes this energy packet when necessary.
- The newly generated ADP is used for yet another round of the phosphorylation procedure to produce ATP, the most common activated carrier in cells. This produces energy for several pumps that actively transfer materials into or out of the cell, establishing an ATP cycle in the cell.

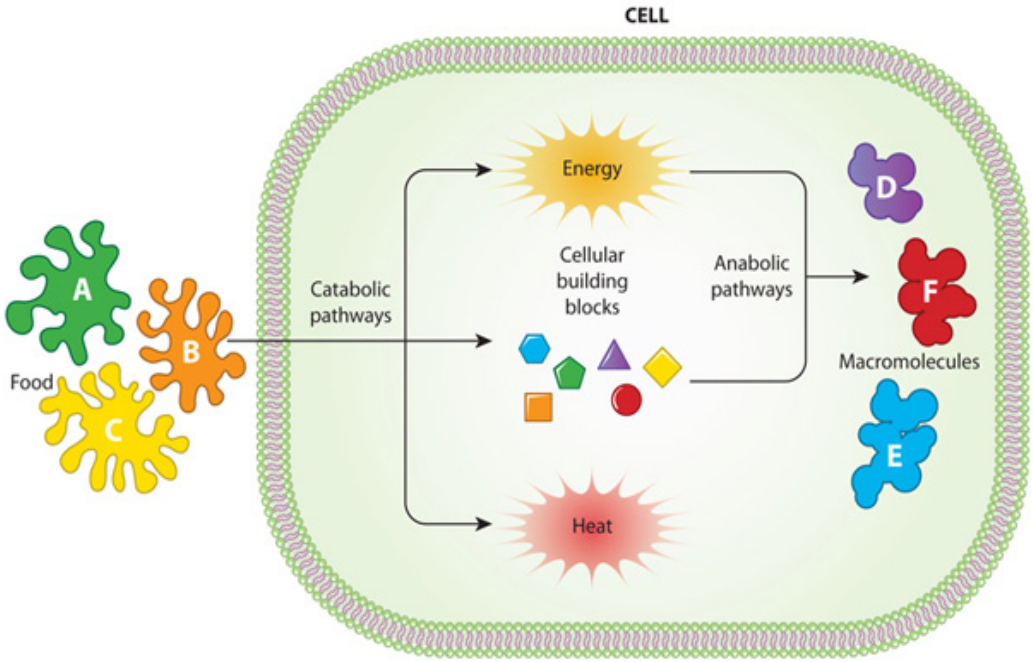
- The products are more stable than the reactants in hydrolysis processes with substantial, negative, standard free-energy shifts for one or more of the following reasons:

(1) Charge separation, as in the case of ATP, relieves bond strain in reactants caused by electrostatic repulsion;
(2) The products are stabilized by ionization, as in the case of ATP, acyl phosphates, and thioesters;
(3) The products are stabilized by isomerization (tautomerization), as in the case of PEP;
(4) The products are stabilized by resonance, as in the case of creatine released from phosphocreatine



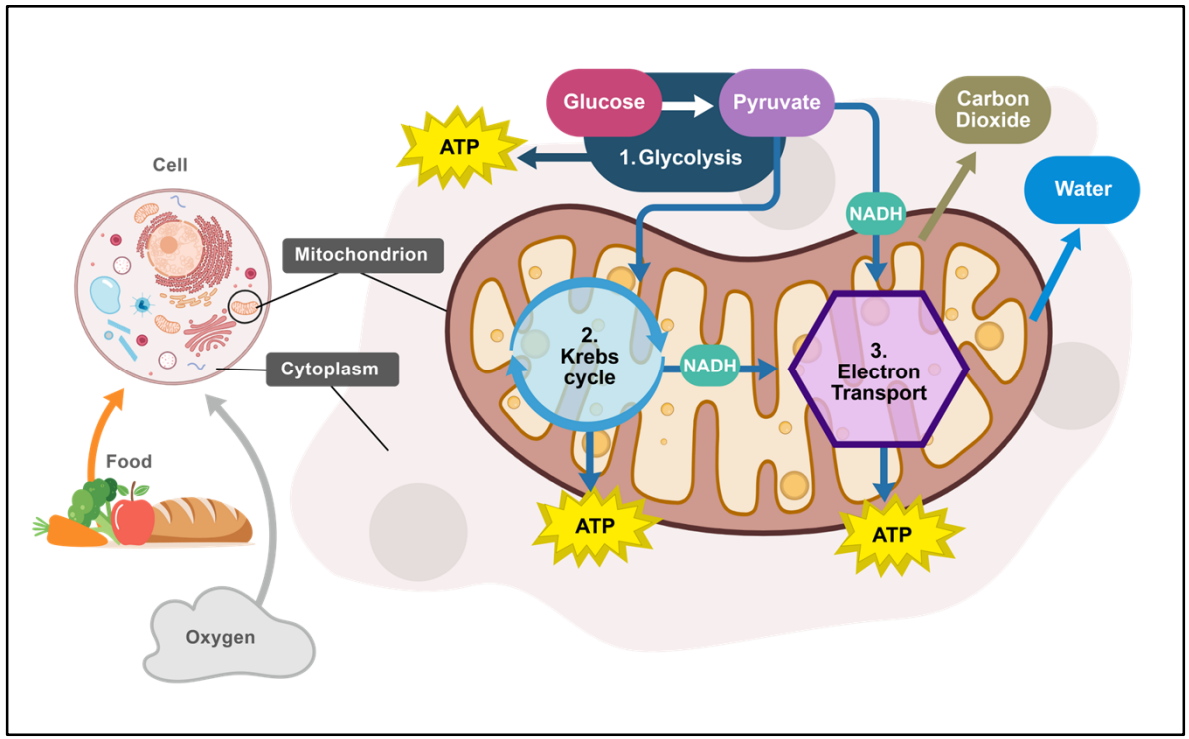
Metabolic pathways

- Metabolism is the **sum of the total chemical activities of all cells** to convert food into energy. Cells must perform an **unending series of chemical reactions** to create the chemicals necessary for the organism to meet its metabolic needs.
- The products of one reaction serve as the starting materials for subsequent ones in these **enzyme-catalyzed** processes, typically **coupled** in series.
- A complex network of **interlinked** reactions is created when the lengthy linear reaction routes, also known as **metabolic pathways**, are connected. Because it enables the cell to precisely control its metabolism, catalysis is advantageous.
- A succession of **metabolic intermediates** called **metabolites** turns the precursor into the final product.
- **Intermediary metabolism** frequently refers to the **sum of all metabolic pathways** that convert **precursors, metabolites, and low-molecular-weight products**.



Aerobic respiration

- **Catabolism** can be defined as the complex organic molecules such as carbohydrates, lipids, and proteins are broken down into smaller, simpler molecules such as lactic acid, CO2, and NH3. **The remaining energy is dissipated as heat**.
- **Anabolism**, known as **the biosynthesis phase**, simple precursors are converted into larger, more complex compounds, including lipids, polysaccharides, proteins & nucleic acids. Anabolic reactions require energy, often provided by the phosphoryl group of **ATP** and the reducing power of **NADH**, **NADPH**, and **FADH2**.

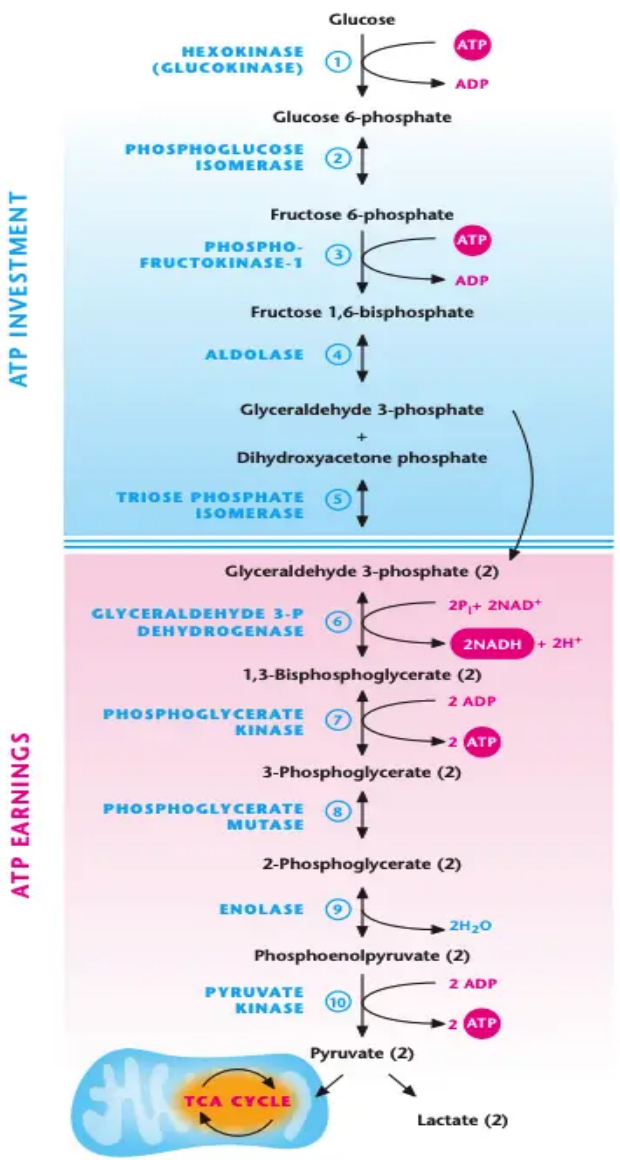


Glycolysis

- Glycolysis is a fundamental metabolic anaerobic pathway responsible for the breakdown of glucose, a six-carbon sugar, into two three-carbon molecules of pyruvate.
- The glycolytic pathway consists of **ten enzymatic reactions**, organized into two main phases: the **energy investment phase** and the **energy payoff phase**. In the investment phase, 2 ATP molecules are consumed to phosphorylate glucose and prepare it for further breakdown. In the payoff phase, 4 ATP molecules and 2 molecules of NADH are produced, resulting in a net gain of 2 ATP molecules per glucose molecule. The overall reaction can be summarized as:

Glucose + 2 NAD^+ + 2 ADP + 2 Pi → 2 Pyruvate + 2 NADH + 2 H^+ + 2 ATP + 2 H_2O.

- Under aerobic conditions, the pyruvate generated enters the mitochondria to undergo further oxidation via the citric acid cycle and oxidative phosphorylation, producing additional ATP.



Glycolysis

- **Key features**:
- **ATP Generation in Anaerobic Conditions**: Glycolysis is the sole pathway for ATP production under anaerobic conditions. It does not require oxygen and occurs in the cytoplasm of cells. This is crucial for cells lacking mitochondria, such as red blood cells, where pyruvate is converted into lactate instead of entering aerobic pathways.
- **Pyruvate Production and Oxygen Dependence**: When oxygen is available, glycolysis produces pyruvate. This pyruvate is then transported into mitochondria, where it enters the TCA cycle (also known as the citric acid cycle or Krebs cycle), contributing to further ATP production.
- **Connection to Anabolic Pathways**: Beyond energy production, intermediates of glycolysis are utilized in anabolic pathways. These pathways help synthesize essential biomolecules, including glycogen, lipids, nucleotides, and proteins. Metabolites from glycolysis can enter various biosynthetic routes such as the pentose phosphate pathway, hexosamine pathway, and the serine and glycerol biosynthetic pathways, supporting cellular growth and function.

Glycolysis

- **<u>Products/Result of Glycolysis</u>:**
- **Glucose is converted to pyruvate**: A single molecule of glucose (6 carbons) is split and oxidized into two molecules of pyruvate (3 carbons each). This step is essential for cells to extract energy from glucose.
- **NAD+ is reduced to NADH:** During glycolysis, NAD+ accepts electrons and is reduced to NADH. This happens particularly during the oxidation of glyceraldehyde 3-phosphate (G3P) to 1,3-bisphosphoglycerate in Step 6. The resulting NADH can later be used in other metabolic processes, like oxidative phosphorylation.
- **ATP is synthesized:** ATP is generated through substrate-level phosphorylation in two distinct steps of glycolysis. First, during Step 7, 1,3-bisphosphoglycerate donates a phosphate group to ADP, forming ATP. Then, in Step 10, phosphoenolpyruvate donates a phosphate group to ADP, forming more ATP. These ATP molecules are directly available for cellular activities.

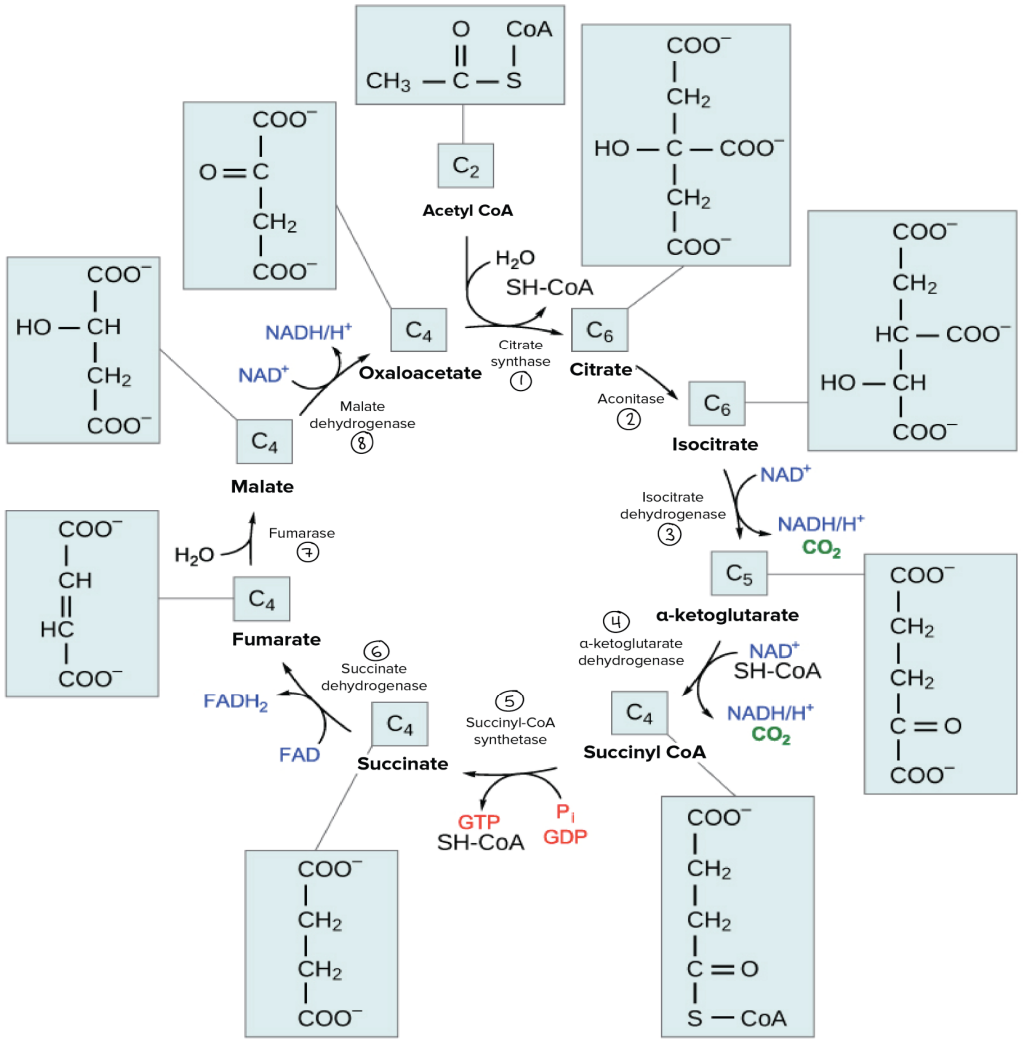
Fates of Pyruvate

- The pyruvate produced by glycolysis is subsequently digested via one of three catabolic processes, except for a few exceptions among bacteria.
- **In aerobic organisms**, glycolysis is the initial step in fully breaking down Glucose. Pyruvate is then oxidized to yield the acetyl group of acetyl-coenzyme A, with loss of CO2; this acetyl group is then completely oxidized to CO2 during the citric acid cycle. The electrons from these oxidations are passed to O2 through a chain of carriers in mitochondria to form H2O. The ATP synthesis in mitochondria is driven by the energy obtained from the electron transfer reactions.
- The **2nd route** for pyruvate is its reduction to lactate via **lactic acid fermentation**. It takes place under **low oxygen** conditions, such as during exercise, the muscle contracts vigorously; during this stage, NADH cannot be re-oxidized to NAD, but NAD is required as **an electron acceptor** for the further oxidation of pyruvate. Under these conditions, pyruvate is reduced to **lactate**, accepting electrons from NADH and thereby regenerating the NAD necessary for glycolysis to continue. Certain tissues and cell types (**retina** and **erythrocytes**, for example) convert glucose to lactate even under aerobic conditions. Lactate is also the product of glycolysis under anaerobic conditions in some microorganisms.
- The **3rd major route** of pyruvate catabolism is **ethanol formation**. In some plant tissues and certain invertebrates, protists, and microorganisms such as **brewer's** or **baker's yeast**, pyruvate is converted to ethanol and CO2 under hypoxic or anaerobic conditions known as ethanol (alcohol) fermentation.

Krebs' Cycle/Tricarboxylic Acid Cycle/Citric Acid Cycle

- This sequence of reactions is also known as the tricarboxylic acid (TCA) cycle, due to 3 carboxyl groups on its first two intermediates, or the Krebs cycle, in honor of its discoverer, Hans Krebs.
- It uses acetyl Co-A formed by the oxidation of glucose-derived pyruvate as its initial substrate and through a sequence of redox reactions, extracts the majority of its bond energy as NADH, FADH2 and ATP molecules.
- The reduced electron carriers NADH and FADH2 produced by the TCA cycle transfer the electron to the Electron transport chain, and by oxidative phosphorylation in cellular respiration, ATP is generated.
- The citric acid cycle occurs in the matrix of the mitochondria in eukaryotes. Whereas in prokaryotes, both of these processes take place in the cytoplasm.

Krebs' Cycle



1. In the initial step of the citric acid cycle, **acetyl Co-A interacts with oxaloacetate**, releasing the Co-A group and producing citrate, **a 6-carbon molecule**.
2. **Citrate is transformed into its isomer, isocitrate**, in the 2nd step. This step involves the **removal and subsequent addition of a water molecule.**
3. **Isocitrate is oxidized**, releasing a CO2 molecule and leaving behind **α-ketoglutarate**, a **5-carbon molecule**. During this stage, **NAD+ is converted to NADH**. Isocitrate dehydrogenase is the enzyme that catalyzes this step and is essential for regulating the rate of the citric acid cycle.
4. **α-ketoglutarate is oxidized**, converting **NAD+ to NADH**, and a CO2 molecule is released. The remaining **4-carbon molecule** combines with Coenzyme A to produce the unstable molecule **succinyl CoA**. α-ketoglutarate dehydrogenase, the enzyme that catalyzes this step, is also essential for regulating the citric acid cycle.

Krebs' Cycle

5. CoA of succinyl CoA is replaced with a phosphate group, which is then transferred to **ADP to produce ATP**. In some of the cells, GDP (guanosine diphosphate) is used in place of ADP to produce GTP (guanosine triphosphate). The **resulting 4-carbon compound** is **succinate**.

6. Succinate is oxidized to generate fumarate, a **4-carbon molecule**. In this reaction, **2 hydrogen atoms and their electrons are transferred to FAD, forming FADH2**. This enzyme is buried within the inner membrane of the mitochondrion, where it allows the direct transfer of electrons from FADH2 to the electron transport chain.

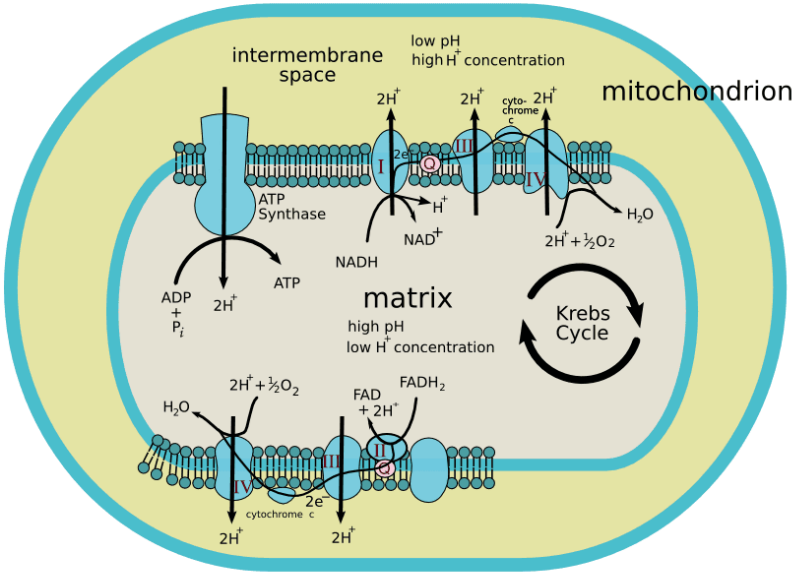
7. Water is added to the 4-carbon fumarate molecule, transforming it into the **4-carbon malate molecule**.

8. Malate is oxidized in the final phase of the citric acid cycle to **oxaloacetate**, leading to the **regeneration of the initial four-carbon molecule**. In this process, another molecule of **NAD+ is converted to NADH.**

Electron Transport Chain

- The electron transport chain is a series of four protein complexes that couple redox reactions, creating an electrochemical gradient that leads to the creation of ATP in a complete system named oxidative phosphorylation. It occurs in mitochondria in both cellular respiration and in chloroplasts for photosynthesis.
- The energy produced in these processes is collected as a **proton gradient** and used to produce ATP via a process known as **chemiosmosis**. The electron transport chain and chemiosmosis work together to form **oxidative phosphorylation**.
- The flow of electrons through the ETC follows a specific sequence: **NADH → Complex I → Ubiquinone → Complex III → Cytochrome c → Complex IV → Oxygen**

Mitochondrial Electron Transport Chain



Electron Transport Chain

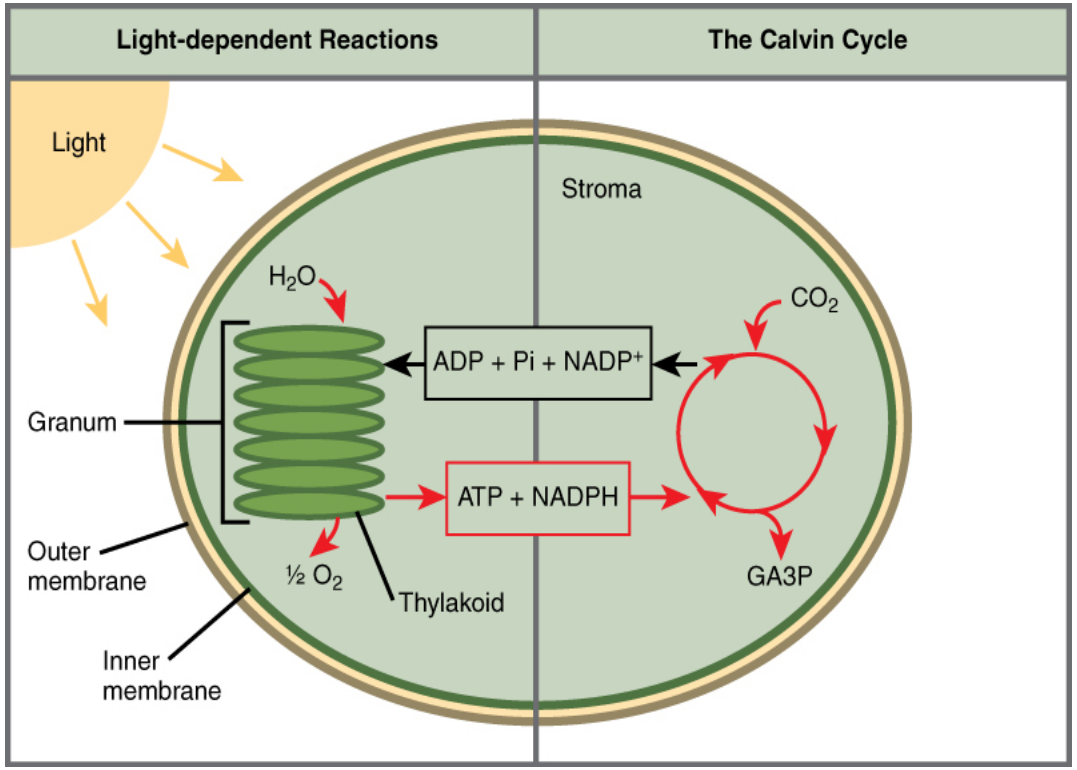
- **Components**
- **Complex I (NADH dehydrogenase)** accepts electrons from NADH and transfers them to ubiquinone. NADH is generated from the oxidation of metabolic substrates (glucose, fatty acids, amino acids) in the citric acid cycle
- **Complex II (succinate dehydrogenase)** accepts electrons from FADH2 and also transfers them to ubiquinone FADH2 is produced during the oxidation of succinate to fumarate in the citric acid cycle
- **Complex III (cytochrome bc1 complex)** accepts electrons from ubiquinone and transfers them to cytochrome c. Ubiquinone, also known as coenzyme Q10, is a lipid-soluble electron carrier that shuttles electrons between complexes I, II, and III
- **Complex IV (cytochrome c oxidase)** accepts electrons from cytochrome c and transfers them to molecular oxygen, the final electron acceptor. Cytochrome c is a small, water-soluble protein that carries electrons from complex III to complex IV. Molecular oxygen is reduced to water (H2O) upon accepting electrons from complex IV

Electron Transport Chain

- **NADH and FADH2 as electron carriers:** Reduced electron carriers (NADH and FADH2), produced in cellular respiration, transfer their electrons to molecules at the beginning of the transport chain. In the process, they turn back into NAD+ and FAD, which can be reused for cellular respiration.
- **Transfer of electron and proton pumping:** As electrons move along the chain, they move from a higher to a lower energy level, releasing energy. Some energy is used to pump H+ ions, moving them out of the matrix and into the intermembrane space. This pumping establishes an electrochemical gradient.
- **Gradient-driven ATP synthesis:** As H+ ions flow down their gradient and back into the matrix, they pass through an enzyme called ATP synthase, which harnesses the flow of protons to synthesize ATP.
- **Water formation by the splitting of oxygen:** At the end of the electron transport chain, electrons are transferred to molecular oxygen, which splits in half and takes up H

Photosynthesis

- Photosynthesis uses solar energy to generate Glucose from carbon dioxide and water. Oxygen gas is the byproduct of photosynthesis. Chlorophyll, carotenoids, and phycobilins are pigments used to absorb solar energy. This light energy is transformed into potential chemical energy during photosynthesis, and Glucose provides the metabolic energy for all cellular functions.
- There are two types of photosynthesis, namely 1) **oxygenic** and 2) **anoxygenic**. Plants, cyanobacteria, and algae use oxygenic photosynthesis, while purple sulfur and green sulfur bacteria engage in anoxygenic photosynthesis. In the case of oxygenic water, oxygen serves as the electron donor, whereas nitrate, hydrogen sulfide, etc., are employed as electron donors in anoxygenic photosynthesis.
- Three turns of the Calvin cycle are required to produce a G3P molecule that can proceed to the production of Glucose. A G3P molecule contains 3 fixed carbon atoms; hence, 2 G3P molecules are required to construct a 6-carbon glucose molecule. Therefore in total 6 rounds of calvin cycle, or 6 CO2, 18 ATP, and 12 NADPH, are required to generate one molecule of Glucose.



Photosynthesis

- **The light-dependent reactions** in the thylakoid membrane require a constant source of light energy. This light energy is transformed into chemical energy by creating two compounds: ATP, an energy storage molecule, and NADPH, a reduced (electron-carrying) electron transporter. During this process, water molecules are also transformed into oxygen gas.
- **Calvin Cycle or C2 cycle** - CO2 is fixed (incorporated into organic molecules) and is used to construct 3-carbon sugars in the Calvin cycle. This process depends on ATP and NADPH, which are derived from photosynthesis. The light reactions occur in the thylakoid membrane, whereas light-independent reactions occur in the stroma (the inner space of chloroplasts).The reactions of the Calvin cycle can be categorized into three distinct phases: **carbon fixation, reduction, and initial molecule regeneration**.

1. **Fixation of carbon** - A CO2 molecule joins with **ribulose-1, 5-bisphosphate**, a 5-carbon molecule (RuBP). This process produces a 6-carbon complex, which is very unstable and therefore breaks apart into 2 molecules of 3-carbon molecules, **3-phosphoglyceric acid (3-PGA)**. This reaction is conducted by the enzyme **rubisco**, also known as **RuBP carboxylase/oxygenase**.
2. **Reduction phase** - 3-PGA is converted into a three-carbon sugar during the reduction phase of the Calvin cycle, which requires ATP and NADPH. This procedure involves two essential steps: (1) Each molecule of 3-PGA obtains a phosphate group from ATP, transforming into **1, 3-bisphosphoglycerate**, a molecule with two phosphate groups. (2) The molecules of 1, 3-bisphosphoglycerate are reduced (gain electrons). Each molecule takes 2 electrons from NADPH and, in turn, loses one of its phosphate groups, transforming into **glyceraldehyde 3-phosphate**, a 3-carbon sugar (G3P). This process generates the byproducts NADP+ and Pi.
3. **Regeneration** - Some G3P molecules are utilized in the synthesis of Glucose, while others are recycled to regenerate the RuBP acceptor. Regeneration is dependent on ATP. It involves a complicated network of chemical processes.

Thank you.